

Experiment No: 11

Title

Write a Python Program for Map Coloring using Constraint Satisfaction Problem (CSP)

Aim

To develop a Python program to solve the **Map Coloring Problem** using the **Constraint Satisfaction Problem (CSP)** approach by assigning different colors to adjacent regions.

Objective

To implement the Map Coloring problem using the backtracking technique of CSP so that no two neighbouring regions are assigned the same color.

Algorithm

Step 1: Start the program.

Step 2: Define the map regions and available colors.

Step 3: Specify the neighbouring regions as constraints.

Step 4: Assign a color to each region one by one.

Step 5: Check whether the assigned color satisfies all constraints.

Step 6: If valid, continue assigning colors to the next region.

Step 7: If no valid color is available, backtrack and try another color.

Step 8: Repeat until all regions are colored.

Step 9: Display the final color assignment.

Step 10: Stop the program.

Program

```
# Map Coloring using CSP (Backtracking)
```

```
regions = ['WA', 'NT', 'SA', 'Q', 'NSW', 'V', 'T']
```

```
colors = ['Red', 'Green', 'Blue']
```

```
neighbours = {  
    'WA': ['NT', 'SA'],  
    'NT': ['WA', 'SA', 'Q'],  
    'SA': ['WA', 'NT', 'Q', 'NSW', 'V'],  
    'Q': ['NT', 'SA', 'NSW'],  
    'NSW': ['SA', 'Q', 'V'],  
    'V': ['SA', 'NSW'],  
    'T': []  
}
```

```
assignment = {}
```

```
def is_safe(region, color):
```

```
    for neighbour in neighbours[region]:
```

```
        if neighbour in assignment and assignment[neighbour] == color:
```

```
            return False
```

```
    return True
```

```
def solve(index):
```

```
    if index == len(regions):
```

```
        return True
```

```
    region = regions[index]
```

```

for color in colors:

    if is_safe(region, color):

        assignment[region] = color

        if solve(index + 1):
            return True

        del assignment[region]

    return False

if solve(0):

    print("\nColor Assignment\n")

    for region in regions:
        print(region, ":", assignment[region])

else:

    print("No Solution Found")

```

Sample Input

No user input.

Sample Output

Color Assignment

WA : Red
NT : Green
SA : Blue
Q : Red
NSW : Green
V : Red
T : Red

Result

Thus, the Python program for **Map Coloring using Constraint Satisfaction Problem (CSP)** was executed successfully, and the map was colored without assigning the same color to any adjacent regions.

